Consistent but Blind? Image Removal Does Not Measure Image Reliance in Medical Vision-Language Models

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Code and data: https://github.com/thaitri2005/Medvlm-Image-Reliance

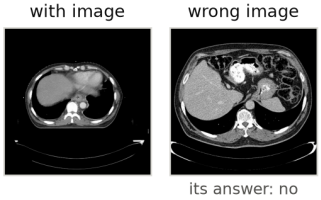
(a) "Is there spinal cord in this picture?" (SLAKE CT, gold: yes)



	with image	2 paraphrases	image removed	quadrant (removed)
MedGemma-4B	yes	yes, yes	yes	Dangerous
Gemma-3-4B	yes	yes, yes	"Please provide the picture."	Ideal

Gemma-3 asks for the image; its forced yes/no choice is "no", so the probe counts it as image-reliant.

(b) "Does the picture contain heart?" (SLAKE CT, gold: yes)



	with image	2 paraphrases	image removed	wrong image	quadrant (removed)	quadrant (wrong)
MedGemma-4B	yes	yes, yes	yes	no	Dangerous	Ideal
Gemma-3-4B	yes	yes, yes	no*	yes	Ideal	Dangerous

** only 51% of first-token probability on yes/no. Same question, opposite verdicts from the two probes.*

Figure 1. **One question, two models, two probes.** (a) Without the image, the medical model MedGemma-4B still answers, and the consistent-but-dangerous taxonomy labels the answer DANGEROUS. Its general-purpose counterpart Gemma-3-4B requests the image; its forced yes/no choice is scored as image-reliant (IDEAL). (b) For the same question, removing the image and substituting a different image (whose correct answer is *no*) yield opposite labels for both models. Actual model outputs; SLAKE CT images (CC-BY-4.0) [17].

Abstract

Whether a medical vision-language model (VLM) uses the image is commonly assessed by removing it: the consistent-but-dangerous taxonomy labels an answer DANGEROUS *when it is stable under paraphrase and unchanged without the image. We ask whether this criterion can rank models. Three medical VLMs and the general-purpose models they were built from (MedGemma-4B and MedGemma-1.5-4B* vs. *Gemma-3-4B; MedVLThinker-3B* vs. *Qwen2.5-VL-3B) answer 3,365 questions on 907 CT, MRI, X-ray and pathology images without the image and under three* in-format *probes that keep an image in the input: a different image, a black image, and an image for which the same question has the opposite answer. The main analysis was pre-registered, and every later probe and model pair was registered before it was run. Under image removal, MedGemma-4B has a 13.7-point [10.0, 17.5] higher* DANGEROUS *share than Gemma-3-4B. No in-format probe corroborates this or any other image-removal difference: of nine model-pair × dataset comparisons, image removal separates the two models in five, and an in-format probe agrees in none. The image-removal differences track refusal (without an image, Gemma-3 declines on 29% of questions, and its forced answers there are unrelated to its answers with the image) and the agreement between a model's priors and the image. In-format probes have a confound of their own: both MedGemma models follow opposite-answer im-*

ages far more than Gemma-3 on SLAKE, which they were trained on (+48.0 and +36.5 points), but not on held-out PathVQA (+0.8 and −5.6). Image removal should therefore not be used to compare image reliance between models; such comparisons require in-format probes on data outside each model's training set.

1. Introduction

A model that answers a question about an image without using the image is not performing visual question answering, however accurate its answers. In general-domain VQA, this concern motivated balanced benchmarks and prior-robust splits [1, 11]. Medical VQA is still evaluated predominantly by accuracy, although the stakes are higher: an answer derived from language priors is most likely to fail on exactly the patients whose images are atypical.

Neither accuracy nor consistency under rephrasing distinguishes using the image from guessing well; a model that ignores the image can be *more* consistent, since no visual signal perturbs it [25]. Sadanandan and Behzadan [24] therefore combined the two criteria per answer. An answer is *consistent* if it is stable under paraphrase, and *image-reliant* if it changes when the image is removed. The resulting quadrants are IDEAL (consistent, reliant), FRAGILE, DANGEROUS (consistent, not reliant) and WORST. On chest X-rays, they found that fine-tuning for consistency moves most answers into DANGEROUS.

Such a taxonomy invites comparisons between models, and these comparisons are only as valid as the reliance criterion. Removing the image changes the input format: a model asked about an absent image may decline, answer from priors, or guess, and none of these responses is equivalent to ignoring an image that is present. For chest radiography, Lotfinia *et al*. [19] accordingly treat performance without the image as a model's default answer rather than as evidence of image use. Whether this matters for comparisons between models, which is how a taxonomy is typically used, has not been tested. We therefore compare image removal with probes that keep an image in the input, on the same questions, for three medical VLMs and the general-purpose models they were derived from: MedGemma-4B [26] and MedGemma-1.5-4B against Gemma-3-4B [10], and MedVLThinker-3B against Qwen2.5-VL-3B, on CT, MRI, X-ray and pathology questions from three public benchmarks.

Figure 2 summarizes the results. Image removal reports differences between medical and general models that no in-format probe corroborates, in every model pair and dataset where it reports a difference. Conversely, the in-format probes indicate substantially greater image reliance for both MedGemma models only on SLAKE, a dataset in their training data; on PathVQA, which none of the medical models was trained on, a pre-registered test finds no such advantage.

Contributions.

- A comparison of four reliance probes on identical questions for three medical/general model pairs and three datasets, showing that no between-model difference found by image removal is corroborated by an in-format probe, and that the largest points the opposite way. The individual probes have precedent [19, 24, 29]; to our knowledge, their conclusions about the same models have not been compared.
- Evidence for two mechanisms behind the image-removal differences: refusal under a forced yes/no format, and agreement between a model's prior and the image content.
- A pre-registered held-out test showing that the in-format advantage of two medical models on their training distribution does not transfer, so that in-format probes also require held-out data.
- Two corrections to the DANGEROUS share: its chance floor on yes/no questions, addressed with Cohen's κ, and a paired decomposition into correct and incorrect DANGEROUS answers.
- An open, pre-registered pipeline covering three datasets, four modalities and five models, including a paraphrase set screened pair by pair for equivalence and the observation that bf16 batching perturbs log-probabilities by up to 1 nat.

2. Related Work

Language priors and modality reliance. Balanced pairs [11] and changing-prior splits [1] showed that general VQA models exploit answer priors. MMStar [3] measures the gain from the image and the leakage of benchmark items into training data, the general-domain counterpart of the exposure we test with held-out data. MM-SHAP [22] and the perceptual score [8] quantify modality contributions by masking or permuting inputs. Contrast sets [7] and NaturalBench [16] score inputs with different answers jointly, which a model that ignores the image cannot pass; our pair accuracy applies this to medical images.

Image use in medical VLMs. PSF-Med [25] measured paraphrase flip rates across medical VLMs on chest X-rays and noted that low flip rates frequently co-occur with high text-only agreement. The quadrant taxonomy [24] defines reliance by image removal and corroborates it with a KL-divergence score, a black image and random image swaps, which agreed with image removal on their models. Zafar *et al*. [29] construct blank, pixel-shuffled, absent and CLIP-retrieved hard-negative images to define reliance metrics

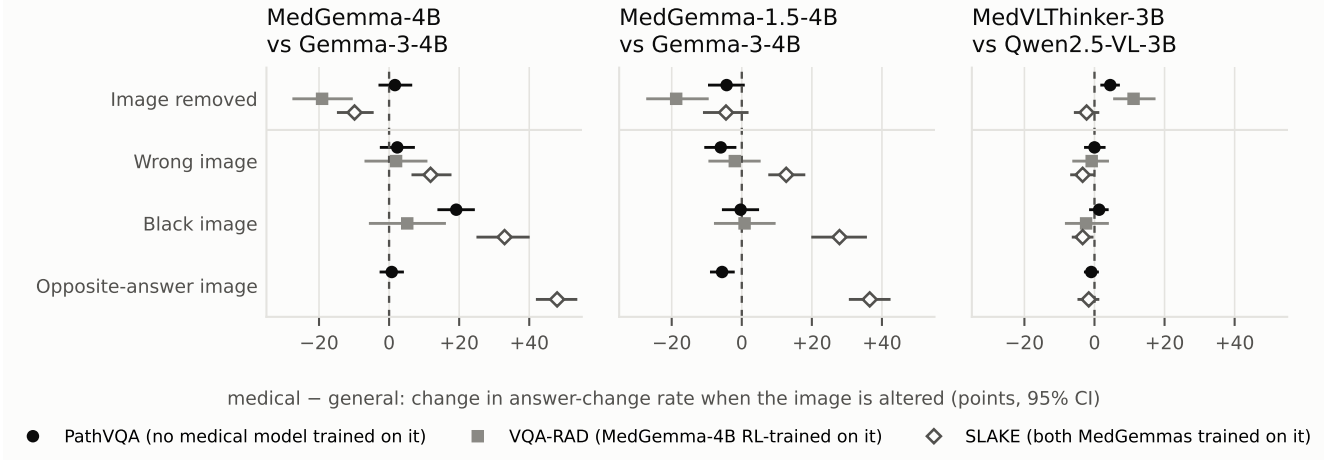


Figure 2. **Whether the medical model appears more image-reliant depends on the probe and on the training data.** Medical − general difference in the answer-change rate (share of yes/no answers that change when the image is altered), by probe (row), dataset (marker) and model pair (panel); 95% CIs from a bootstrap over images. Image removal (top row) separates the two models in five dataset cells; no in-format probe (lower rows) agrees in any of them. The in-format probes separate the MedGemma models from Gemma-3 mainly on SLAKE (open diamonds), part of their training data. On held-out PathVQA (filled circles; registered probe R4), the opposite-answer probe finds no MedGemma advantage.

and train a model against them. Lotfinia *et al*. [19] swap, occlude and remove chest radiographs, take the opposite-label swap as the primary test because accuracy without the image reflects a model's default answer and the label distribution, and find that model class (medical or general) does not predict image use. ProbMed [28] pairs medical questions with adversarial counterparts and finds large models below chance. Pathology VLMs reach high accuracy without the image [31], and medical MLLMs frequently attend to irrelevant regions [18]. Grounding inputs can even reduce closed-ended accuracy [4]; HEAL-MedVQA [20] and MedHEval [2] benchmark localization and hallucination. These studies use a single probe, treat several as interchangeable, or argue against removal on conceptual grounds; we measure how the conclusions of different probes about the same models diverge, explain the divergence, and test it inside and outside the models' training data.

Answering without evidence. MedVIGIL [14] tests whether models abstain when visual evidence is corrupted, and Vohra [27] reports that frontier VLMs produce diagnoses for images that were never provided, and attaching even a blank image shifts refusal behavior [32], so neither an absent nor a blank image is a neutral control. Four of our five open models decline almost always when permitted to; we show how enforcing an answer affects a reliance criterion.

Selective prediction. Ranking answers by a score and abstaining on the lowest-ranked ones is evaluated by AUROC and risk–coverage [9]; we use this framework to test whether image reliance identifies errors.

3. Method

3.1. Data

SLAKE [17] (English; CT, MRI, X-ray; 2,114 test and validation questions; the only dataset with modality labels, with the validation split included so that each modality has 53–82 images). **VQA-RAD** [15] (radiology, clinician-written; all 451 test questions from the original release). **PathVQA** [12] (pathology, generated from textbook captions; 600 yes/no and 200 open test questions sampled with a fixed seed). A question is yes/no if its reference answer is *yes* or *no*. The **primary analysis set** comprises the 1,206 yes/no *test* questions on 670 images. Probe R4 uses a separate set: all 1,192 PathVQA test and validation yes/no questions that are also asked of another image with the opposite answer (682 images). Sampling, partner images and image hashes are fixed in committed manifests, so that every run receives identical inputs.

3.2. Models

MedGemma-4B (`google/medgemma-4b-it`) is an open medical VLM whose SigLIP-based encoder [30] and language model were further trained on medical data. Its technical report [26] lists SLAKE (450 items, the size of SLAKE's training split) and VQA-RAD (reinforcement-learning stage) among its training data and states that PathVQA was excluded. **Gemma-3-4B** (`google/gemma-3-4b-it`) has

the same architecture and size. **MedGemma-1.5-4B** (`google/medgemma-1.5-4b-it`, released in 2026; its model card lists SLAKE) is compared with the same Gemma-3-4B, and **MedVLThinker-3B** (`UCSC-VLAA/MedVLThinker-3B-RL_PMC`, Qwen2.5-VL-3B-Instruct trained with reinforcement learning on PMC-VQA) with its base model **Qwen2.5-VL-3B-Instruct**. A medical model and its base differ in their entire post-training, not only in medical data; we therefore compare *checkpoints* rather than attribute differences to medical training as such. All five models run unquantized in bf16 on a single 16 GB GPU, since quantization would perturb the probabilities under study, and receive identical prompts: a fixed system message, the question, and "Answer with only 'yes' or 'no'." or "Answer with a single word or short phrase." Each model added after the primary pair passed a registered format check (mean first-token yes/no probability mass with the image ≥ 0.9 in every dataset of a 330-question pilot; observed 0.93–0.999).

3.3. Probes

Each question is posed with its own image (**orig**) and as screened paraphrases with the same image (**para**$_k$; Sec. 3.5). Image reliance is probed by altering only the image or the instruction:

- **Image removed** (*noimg*): image tokens omitted, text unchanged; the reliance criterion of the taxonomy [24].
- **Wrong image** (*shuf*): another image from the same dataset and modality, selected by a hash of the question identifier.
- **R1, black image**: a black image in place of the original, preserving the input format but removing the content.
- **R2, abstention permitted**: image removed, with the instruction extended by "If this cannot be determined without seeing the image, answer 'unknown'."
- **R3, opposite-answer image**: another image of the same dataset and modality for which the *same* question has the opposite reference answer; a model that uses the image must change its answer. Partners exist for 367 SLAKE yes/no questions (130 images), whose templated questions recur across images.
- **R4**: R3 applied to the 1,192 PathVQA questions above, outside the listed training data of every medical model.

We refer to the wrong-image, black-image and opposite-answer probes as *in-format* probes, since an image is present in the input. They are not equally informative. Only the opposite-answer image fixes what an image-reliant model must do, namely change its answer, and change it to the correct one; a black image is not a neutral input [32], and a random image often has the same correct answer. We therefore treat R3 and R4 as the reference and report the other probes alongside them. The first two probes belong to the pre-registered main run; R1–R4 and the two additional model pairs were each registered after the results that motivated them and before they were run.

3.4. Measures

Yes/no answers are obtained from a single forward pass, $P(\text{yes}) = \sigma(\text{LSE}_{t\in Y}\,\ell_t - \text{LSE}_{t\in N}\,\ell_t)$, over the single-token variants of *yes* and *no*; the first-token mass $m = P(Y) + P(N)$ is recorded as a format check. **Open answers** are decoded greedily and scored by token-F1 ≥ 0.5 after normalization, with a guard that prevents opposite terms (left/right, upper/lower, *etc*.) from matching. An answer is **consistent** if the answers to the first two valid paraphrases equal the *orig* answer, and **reliant** under a probe if the probe answer differs from the *orig* answer; the share of reliant answers is the **answer-change rate** of that probe. The **quadrant** follows from (consistent, reliant). **Refusal** is $m < 0.5$, *i.e*. the majority of first-token mass falls on neither *yes* nor *no*; free generation confirms that these responses are requests for the image (Fig. 1a). **Pair accuracy** (R3, R4) is the share of swaps answered correctly on both images, which is zero by construction for a model that ignores the image. The **image PMI** of an answer $\hat{y}$ is $\log p(\hat{y} \mid I, q) - \log p(\hat{y} \mid \varnothing, q)$. **Chance correction** uses Cohen's κ [5] between *orig* and probe answers; the *chance floor* of the DANGEROUS share is $P(\text{consistent}) \cdot p_e$, where p_e is the agreement expected from the two answer distributions alone.

3.5. Paraphrase construction and screening

Paraphrase consistency is meaningful only if a paraphrase cannot change the correct answer. Candidates are generated by Qwen2.5-3B-Instruct [23] (greedy decoding; five styles) and by two deterministic templates that copy every content word ("Is there X?"→"Is X present?"). A rule-based validator rejects changes to laterality, negation and absence terms, quantities, antonyms, imaging planes, question form, answer leakage, near-copies and length. Two audit rounds accepted 81% and 83% of sampled pairs, below the pre-specified 85%; we therefore tightened the rules, excluded a "plain language" style, and **judged every unique rule-valid (question, paraphrase) pair**, 4,890 in total, against a written rubric (Appendix B). The judge was an LLM (Claude); see Sec. 6. It rejected 1,208 pairs, each with a recorded reason (*e.g. mass→lesion*, *is enlarged→can be enlarged*); unjudged pairs are rejected by construction. Of the primary-set questions, 84% retain at least two valid paraphrases; quadrant analyses use these (1,012 questions, 600 images).

3.6. Statistics and pre-registration

Hypotheses, endpoints, tests and exclusions were fixed before the main run [21]; every later change is listed in Ap-

pendix C, and post-hoc analyses are identified as such. Because several questions share an image, all intervals are percentile 95% CIs from a cluster bootstrap over images [6] ($B = 2000$), with two-sided bootstrap p-values. Holm correction [13] is applied within four families (Tab. 2): **P1a–c** (accuracy without the image minus the majority-answer prior) and **P2** (AUROC of image PMI minus AUROC of confidence); **H6–H8** (probes R1–R3, MedGemma − Gemma-3); **H9–H11** (additional pairs, medical − general: more frequent following of R3 images, fewer answer changes without the image, less refusal); and **H12a–c** (R4, with a decision rule fixed in advance). **P3**, the paired difference in DANGEROUS share, forms a separate family. A finding is considered robust only if its sign holds across a grid of F1 thresholds (0.3/0.5/0.8), paraphrase counts (1/2/3) and "strong" flips.

4. Results

All runs completed: 18,896 model calls per model in the main run, 3,535 for R1–R3 and 2,384 for R4, for each of the five models. The pipeline reproduces MedGemma's published SLAKE score (token-F1 70.2 [67.5, 72.7] *vs*. 72.3 on the model card; Gemma-3: 45.1 *vs*. 40.2), and the yes/no format mass with the image is ≥ 0.997 in every dataset. With the image, MedGemma-4B is more accurate than Gemma-3 on yes/no questions by 18 points on SLAKE and 25 on VQA-RAD, both part of its training data, and by 2 points on PathVQA, which is not (Tab. 1).

4.1. Under image removal, the medical models appear less image-reliant

Under image removal, MedGemma-4B's DANGEROUS share is 50.6/46.8/56.5% (PathVQA/SLAKE/VQA-RAD), compared with 43.2/28.1/31.0% for Gemma-3 (Fig. 3, left). The paired difference, **P3** $= +13.7$ **points** [10.0, 17.5] ($p < 0.001$; 1,012 questions, 600 images), is positive in every dataset (PathVQA +7.5 [2.6, 12.3], SLAKE +18.6 [12.2, 24.9], VQA-RAD +25.6 [14.7, 36.5]) and in every cell of the sensitivity grid. MedGemma-4B is also the more consistent model (paraphrase flip rate 6.8% *vs*. 13.6% on SLAKE), and MedGemma-1.5 shows the same pattern (+9.8 [5.8, 13.9]). Taken at face value, the medical models more often give stable, confident answers that do not depend on the image.

4.2. No in-format probe corroborates image removal

Figure 2 reports the answer-change rate of every probe by model pair and dataset. Image removal separates the medical model from its base in five of the nine cells: lower change rates for both MedGemma models on VQA-RAD (-19.1 [$-27.6, -10.4$] and -18.7 [$-27.3, -9.4$]) and for MedGemma-4B on SLAKE (-9.9 [$-14.9, -4.4$]), and higher rates for MedVLThinker on PathVQA and VQA-RAD (+4.5 [1.7, 7.2] and +11.2 [5.3, 17.4]). In four of these cells, no in-format probe detects a difference, and their intervals exclude a difference as large as the one image removal reports (a statement about effect size, not merely a non-significant test; this analysis is post hoc and the nine cells are not corrected for multiplicity); in the fifth, MedGemma-4B on SLAKE, every in-format probe shows a difference in the opposite direction (wrong image +11.8, black image +33.0, opposite-answer image +48.0 points). Pooled over datasets, the primary pair shows the same divergence: image removal -6.1 [$-9.5, -2.7$], wrong image +5.1 [1.7, 8.5], black image +20.3 [15.9, 24.6]. The two main-run probes also agree poorly at the question level (κ between their reliance labels: -0.11 to 0.23), and the modality ordering of the DANGEROUS share reverses between them, partly because a randomly chosen partner image often has the same correct answer (all 23 chest X-ray partners that pose the same question).

In-format probes on SLAKE. The registered probes R1 and R3 were run on the primary pair after the main results. H6, which predicted that MedGemma-4B's answers would be more stable under a black image, was not supported; the effect was in the opposite direction ($\kappa(\text{image}, \text{black}) = 0.11$ *vs*. 0.33). On a black image, MedGemma-4B's yes-rate falls from 58% to 28%, consistent with reading an empty image as an absence of findings, whereas Gemma-3's rises from 64% to 84%. H8 was supported: on SLAKE's opposite-answer swaps, MedGemma-4B follows the image in 72.8% of cases and Gemma-3 in 24.8%, and MedGemma-4B answers both images correctly in 71.7% of swaps, compared with 23.4%. MedGemma-1.5 shows the same pattern on SLAKE (H9, +36.5 [30.6, 42.1]); MedVLThinker does not differ from its base under any in-format probe on any dataset, apart from a small black-image difference on SLAKE (-3.4 [$-6.5, -0.3$]). Section 4.4 tests whether the MedGemma advantage extends beyond SLAKE.

4.3. Sources of the image-removal differences

Refusal. The pre-registered format check ($m \geq 0.9$) failed in a single condition, image removal. Free generation shows why: "*Please provide the picture. I need to see the image to determine if there is a spinal cord present*" (Gemma-3). Without the image, Gemma-3 declines on 354 of the 1,206 questions, MedGemma-4B on 219 and MedGemma-1.5 on 8 (Fig. 4); MedVLThinker and Qwen2.5-VL never decline. On declined questions, the forced answer is predominantly *no* and unrelated to the model's answer with the image ($\kappa = -0.01$ for Gemma-3, 0.04 for MedGemma-4B), so 53–55% of these answers are classified as image-reliant. The largest image-removal

Table 1. Accuracy (%) on test questions. Yes/no: with the image, without it ("none") and with a different image, alongside the majority-answer prior. Open: token-F1 ≥ 0.5 with the image. CIs are reported in the repository.

		MedGemma-4B (yes/no)			Gemma-3-4B (yes/no)			Open, with image	
Dataset	Prior	image	none	wrong	image	none	wrong	MedGemma	Gemma-3
PathVQA (pathology)	52.2	65.5	57.2	51.0	63.2	53.5	46.7	10.5	4.0
SLAKE (CT/MRI/X-ray)	50.7	**86.5**	47.6	67.0	68.7	47.6	62.3	66.7	37.1
VQA-RAD (radiology)	53.0	**81.3**	60.6	70.5	56.2	55.4	60.2	46.0	33.5

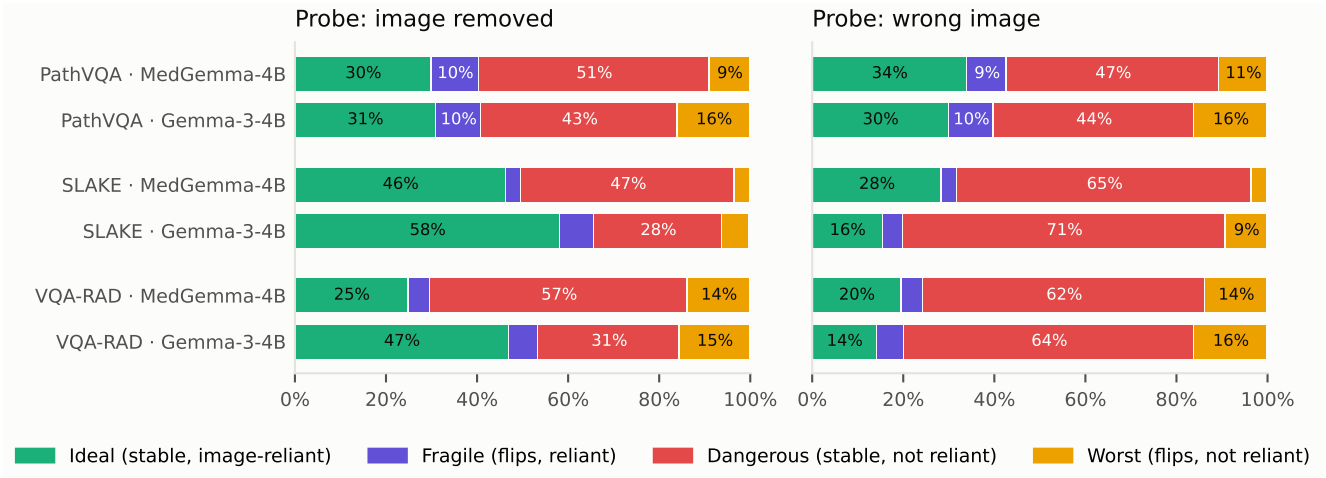


Figure 3. Quadrant composition of yes/no answers (test split), MedGemma-4B and Gemma-3-4B. Left: reliance assessed by removing the image (the taxonomy's definition). Right: by a different image of the same modality. Shares are uncorrected; on yes/no questions they include chance agreement (Sec. 4.5) and correct answers (Sec. 4.3).

Table 2. Confirmatory tests. P1–P2 concern MedGemma-4B (P2 on the model's own answer, Appendix C); all other tests are paired differences (points, or κ for H6). Pairs: MedGemma-4B *vs.* Gemma-3 (P3, H6–H8, H12a); A: MedGemma-1.5 *vs.* Gemma-3 (H12b); B: MedVLThinker *vs.* Qwen2.5-VL (H12c). 95% CIs from a bootstrap over images.

	Test	Estimate	Outcome
Fixed before the main run			
P1a	blind − prior, PathVQA	+5.0 [0.9, 8.8]	p_{Holm}=.042
P1b	SLAKE	−3.1 [−7.8, 1.8]	n.s.
P1c	VQA-RAD	+7.6 [0.4, 14.5]	p_{Holm}=.072
P2	AUROC: PMI − conf.	−0.20 [−0.25, −0.15]	lower
P3	DANGEROUS, image removed	+13.7 [10.0, 17.5]	p<.001
Registered later, each before it was run			
H6	κ(image, black) > 0	−0.22 [−0.30, −0.14]	reversed
H7	declines, R2 < 0	−6.7 [−8.2, −5.3]	supported
H8	follows R3 > 0	+48.0 [41.9, 53.8]	supported
H9	A / B: follows R3 > 0	+36.5 / −1.6	A only
H10	A / B: removed < 0	−7.4 / +3.9	A only
H11	A / B: refuses < 0	−28.7 / 0.0	A only
H12a	R4 (PathVQA) > 0	+0.8 [−2.7, 4.2]	n.s.
H12b	A: R4 > 0	−5.6 [−9.1, −2.0]	reversed
H12c	B: R4 within ±5	−0.9 [−3.0, 1.3]	supported

differences occur where refusal rates differ most: on VQA-RAD, Gemma-3 declines on half of the questions, 23 points more often than MedGemma-4B and 48 points more often than MedGemma-1.5. When abstention is permitted (R2), MedGemma-4B declines on 93.0% of questions and Gemma-3 on 99.7% (H7, −6.7), eliminating most of the 11.2-point difference observed under the forced format. For MedGemma-1.5, which almost never declines, the DANGEROUS difference falls to +3.0 [−3.6, 9.5] on the 467 questions that both models answer.

Agreement between prior and image. Refusal does not account for all of the difference: on the 436 questions that both MedGemma-4B and Gemma-3 answer without the image ($m \geq 0.9$), P3 remains +8.0 [2.9, 13.3], and image removal still separates the two Qwen-based models, neither of which declines. When a model's prior matches the image content, its answer is the same with and without the image, and image removal classifies the answer as image-independent. MedGemma-4B's answers without the image exceed the majority-answer prior on PathVQA, which it was not trained on (P1a, +5.0; +11.9 [7.1, 16.6] among questions it answers), and agree with its answers

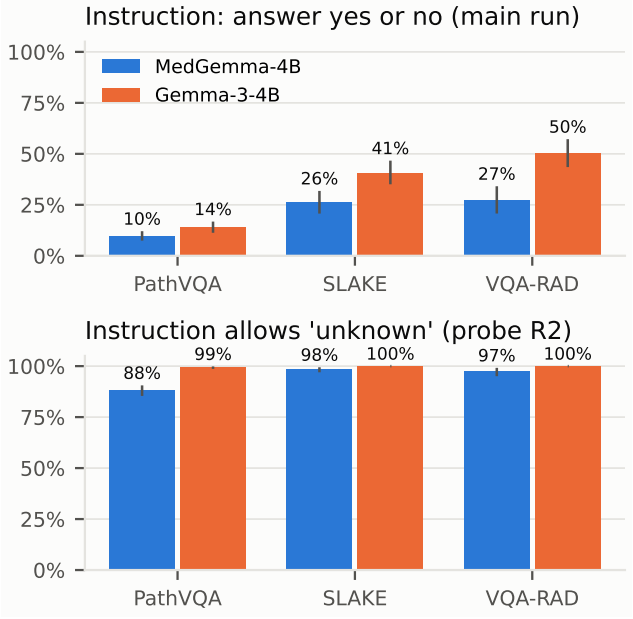


Figure 4. Share of yes/no questions on which the model declines to answer when the image is removed ($m < 0.5$; 95% CIs from a bootstrap over images). Top: the main-run instruction ("Answer with only yes or no"). Bottom: probe R2, whose instruction permits "unknown".

with the image beyond chance ($\kappa = 0.21$ *vs.* 0.11 for Gemma-3). In the Qwen pair, the base model's answers without the image agree more ($\kappa = 0.16$ *vs.* 0.09), and image removal correspondingly classifies the base model as less reliant. The correctness decomposition shows the composition of MedGemma-4B's excess (Fig. 5, bottom): DANGEROUS-and-correct is 39.2% *vs.* 25.1% (+14.1 [11.1, 17.1]), whereas DANGEROUS-and-incorrect, the category that would mislead a user, is 11.3% *vs.* 11.7% (-0.4 [-2.7, 2.0]). MedGemma-1.5's excess is likewise correct (+9.3 correct, +0.5 incorrect).

4.4. Held-out data (probe R4)

Both MedGemma models were trained on SLAKE (the evaluated images are disjoint from its training split, but the image distribution and question templates are shared), whereas MedVLThinker was not. Training exposure therefore explains the SLAKE results as well as greater image reliance does. R4 repeats the opposite-answer probe for all five models on PathVQA, which was excluded from MedGemma's training (Tab. 2). Every model follows the partner image in 43–49% of the 1,192 swaps and answers both images correctly in 36–43%, far above the 0% of a model that ignores the image: all five models use pathology images, to a similar degree. MedGemma-4B does not differ from Gemma-3 (H12a: +0.8 [-2.7, 4.2]), MedGemma-1.5 follows the image *less* often (H12b: -5.6 [-9.1, -2.0],

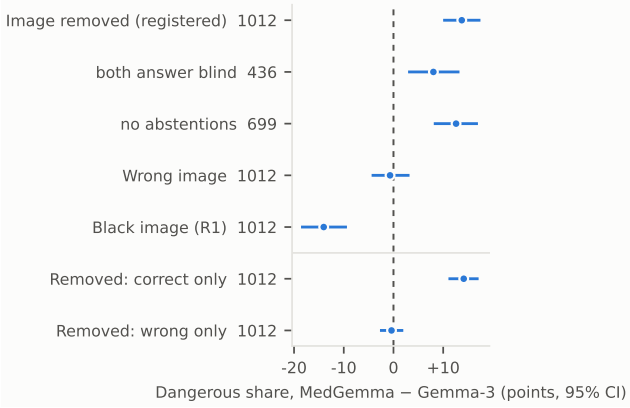


Figure 5. MedGemma-4B − Gemma-3 difference in DANGEROUS share under each probe and subset, and decomposed by whether the DANGEROUS answer is correct (paired; 95% CIs from a bootstrap over images).

$p_{\text{Holm}} = 0.003$), and MedVLThinker matches its base as predicted (H12c). Resampling PathVQA's 154 question templates instead of images leads to the same conclusions. Under the decision rule fixed before the run, the SLAKE difference is reported as possibly reflecting the training distribution (H12a), and the claim of greater image reliance is withdrawn for MedGemma-1.5 (H12b). R4 cannot separate training exposure from a difference between domains. It is not, however, a domain new to MedGemma: its training included 32.5 million histopathology patch–caption pairs [26]; only the PathVQA questions and images are held out. Despite this training, the advantage seen on SLAKE does not appear. Accuracy shows the same pattern: MedGemma-4B's advantage over Gemma-3 is 18 points on SLAKE yes/no questions and 2 points on PathVQA (Tab. 1). On PathVQA, only the black-image probe separates MedGemma-4B from Gemma-3 (+19.2), consistent with its response to an empty image rather than with greater use of image content; the content-based probes (different image, opposite-answer image) show no difference.

4.5. The chance floor

An answer is classified as image-independent whenever the answer without the image equals the answer with it, which on yes/no questions frequently occurs by chance. On the P3 set, MedGemma-4B's answers are unchanged without the image in 58.6% of cases, against 49.1% expected from its answer distributions alone (Gemma-3: 49.7% *vs.* 45.7%). Combined with consistency, this yields a **chance floor of 41.5% (MedGemma-4B) and 35.8% (Gemma-3)**, against observed DANGEROUS shares of 50.5% and 36.8%. Approximately 6 of the 13.7 points of P3 reflect the difference in floors, partly because MedGemma-4B is more consistent (84.5% *vs.* 78.4%), a desirable property that the met-

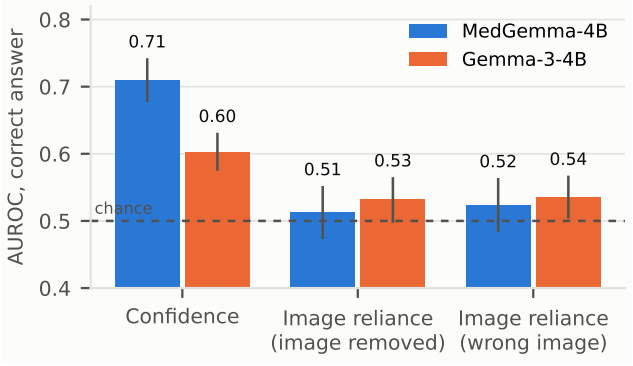


Figure 6. AUROC for ranking correct above incorrect yes/no answers (test split, 95% CIs): the model's confidence *vs*. label-free image reliance under each probe.

ric counts against it. The chance-corrected difference is $\Delta\kappa = +0.112\,[0.044, 0.180]$.

4.6. Image reliance does not identify errors

If image-independent answers were the unreliable ones, image reliance could serve as an error signal. Ranking yes/no answers by image PMI separates correct from incorrect answers at chance level (Fig. 6; AUROC **0.513** [0.473, 0.552] for MedGemma-4B, 0.532 for Gemma-3), whereas the model's own confidence reaches 0.709 and 0.603 (P2: higher by 0.197 [0.145, 0.250] and 0.071 [0.026, 0.116]). The degree to which an answer depends on the image is thus not informative about its correctness.

5. Discussion

What image removal measures. Removing the image poses a question about an image that is absent. The response depends first on whether the model answers at all and on the quality of its priors, and only then on whether it would have used the image. Across three model pairs and three datasets, every difference that image removal reported between a medical model and its base was uncorroborated by the in-format probes, and the largest differences coincided with the largest differences in refusal. The corroborating checks of the original study (black image, random swaps) agreed with image removal on its chest X-ray models [24]; on our models and data they do not, in line with the conceptual argument against removal for chest radiography [19]. A DANGEROUS share is therefore underspecified without its probe, and a comparison of DANGEROUS shares obtained by image removal may reflect refusal policy more than image use.

What the in-format probes measure. The in-format probes are not free of confounds either. On SLAKE, they indicate that both MedGemma models rely on the image far more than Gemma-3; on PathVQA, outside their training data, they do not. Image reliance is a property of a model *and* a data distribution, and in-format probes evaluated on a model's training distribution may reflect what the model learned there rather than how it uses images in general. On opposite-answer swaps, following the image and answering correctly largely coincide: pair accuracy is 83–99% of the follow rate for every model and set, so the probe measures the *correct* use of the image, which is the property that matters for a diagnostic answer, rather than sensitivity to arbitrary changes. For the three pairs studied, no general claim of greater image reliance by the medical model survives evaluation on held-out data.

Residual risk. MedGemma-4B's additional DANGEROUS answers are correct, and the share of stable, image-independent and incorrect answers is the same for both models. MedGemma-4B makes fewer errors overall (25.7% *vs*. 37.6%), but a larger share of them fall into this category (44% *vs*. 31%, +12.9 [6.0, 19.9]); such errors would not be detected by consistency or reliance checks.

Recommendations for reliance audits. (1) Assess reliance with in-format probes, preferably with an image that requires a different answer, and report the agreement between probes. (2) Evaluate in-format probes on data outside each model's training set, and consult training documentation before comparing models. (3) Report the DANGEROUS share together with its chance floor, or as κ; on balanced yes/no data, a model whose answers without the image are pure noise is about 50% "not reliant". (4) Decompose DANGEROUS answers by correctness. (5) Permit abstention, and treat declining without an image as appropriate behavior rather than as a format failure.

6. Limitations

Three model pairs from two families establish that image removal *can* disagree with in-format probes and indicate why; they do not establish how often this occurs across models. The refusal and prior-agreement accounts and the per-dataset analysis (Fig. 2) were formulated after the primary results; among the dataset-specific comparisons, only R4 was registered. R4 covers a single modality, pathology, on which all models perform modestly (pair accuracy 36–43%); it argues against a general advantage but cannot exclude one in every domain. R3 and the SLAKE results rest on templated questions that recur across images; resampling templates instead of images leaves the conclusions unchanged. MedGemma-4B was also trained on VQA-RAD, nearly all of whose test images appear in its training split, so its VQA-RAD results may reflect exposure; the training data of the general models are not published. A medical

model and its base differ in their entire post-training, not only in medical data. Refusal is inferred from first-token probability mass and confirmed on a sample by free generation. All models answer in a forced format from the first token; MedVLThinker was trained to reason before answering, so its results describe the model without its reasoning step. A single instruction template was used, and refusal depends on the instruction (R2). The opposite-answer pairs inherit the label noise of their datasets (PathVQA's labels are generated automatically), which lowers attainable pair accuracy equally for both models of a pair. Only open models of 3–4B parameters were studied; larger and proprietary models may behave differently. Paraphrase equivalence was judged by an LLM and **was not validated by independent human raters**; a check of 50 paraphrases, rated by the author with AI assistance, accepted 88% (above the pre-specified 85%) but is not independent of AI judgment. This is the principal limitation of the consistency measure; the probe comparison does not use paraphrases. PathVQA questions are generated automatically and are often ungrammatical. The chance correction and the correctness and refusal decompositions were introduced after the results were known. Excluding the 330 pilot questions used to tune the design leaves every conclusion unchanged (P1a +6.3 [2.4, 10.6], P3 +12.9 [8.9, 17.1]). Benchmark VQA is not clinical use, and we make no clinical claims.

7. Conclusion

By the image-removal criterion, MedGemma-4B gives image-independent answers more often than its general-purpose counterpart. No probe that keeps an image in the input corroborates this difference, in any model pair or dataset we examined. Image removal measures whether a model answers without an image and whether its priors agree with the image, rather than how much it relies on the image; probes that keep the image indicate greater reliance for the medical models only on data they were trained on. Comparisons of image reliance between models therefore require in-format probes, preferably images that call for a different answer, on held-out data, together with chance correction and a decomposition by correctness.

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A. Implementation details

Pipeline. The experiments are implemented as a Python package (`medvlm`) with a command-line interface: `prepare` (download, deterministic sampling, hash-verified images), `paraphrase` (generation with rule-based and judgment-based validation, fail-closed), `pipeline` (resumable inference in parts with checkpoint analysis), `probes` (R1–R3), `contrast` (R4), and `analyze`, `compare`, `posthoc`, `replication` and `contamination`, which produce all tables and figures in this paper. Configurations are layered by base, profile, model and hardware. Each job is a (question, condition) pair keyed by identifier, so that interrupted runs resume without recomputation; every run records its configuration, commit and library versions. Registrations are git tags.

Inference. PyTorch 2.11 and Hugging Face Transformers 5.19, bf16, SDPA attention, one RTX 4080 SUPER (16 GB). MedGemma-4B requires 0.23 s per call (18,896 calls in about 75 minutes, 8.2 GB peak memory). During pre-flight testing, we found that in bf16 the *batch size* alone changes teacher-forced log-probabilities by up to 1.09 nats (median 0.17) on identical, unpadded inputs, while $P(\text{yes})$ changes by at most 0.008. An fp32 output head did not remove the effect, which therefore originates inside the network. All runs consequently use batch size 1, at a cost of about 1.5× in throughput. A regression check reproduced the pilot run, performed on a T4, on the final GPU (98.5% identical answers, mean $|\Delta P(\text{yes})| = 0.004$).

Testing. 103 unit tests and 7 tests with real models cover, among others, scoring edge cases, the paraphrase rules (including every failure class found in the audit), batching equivalence, crash-safe resumption, the coverage of the cluster bootstrap (by simulation), the test-split restriction of

P3, the chance correction of κ, the construction of opposite-answer partners, and an oracle-*vs*.-blind check of pair accuracy.

B. Paraphrase equivalence rubric

A pair is **accepted** only if both questions ask exactly the same thing, so that the correct answer cannot differ. Permitted: perception framing ("can X be seen", "is X visible", "can X be seen to be enlarged"), exact synonyms (clavicle/collarbone, encephalon/brain, intravenous contrast/dye), and reordering. **Rejected**: a statement of fact turned into one of possibility or capability ("Is the heart enlarged?"→"Can the heart be enlarged?"); a broadened or narrowed term (mass→lesion, organs→structures, digestive system→GI tract, skull fracture→cranial injury); a changed scope or presupposition ("Do the organs exist in X"→"Are there organs in X"); added qualifiers, findings or modalities ("bilaterally", "on imaging studies", "radiograph"); a reversed or altered relation ("does A show B"→"is A associated with B", "mass lateral to X"→"X lateral to mass"); antonyms (healthy↔unhealthy, coronal→transverse); two questions in one; and malformed text. Rejections by class: relation 447, term 364, modality of statement 106, added content 102, organ→structure 47, generic collapse 30, polarity 19, scope 9 (plus rejections of the template-based pass).

C. Deviations from the pre-registration

All changes made after the analysis plan was fixed, with their rationale.

Change	Rationale and effect
Batch size 1 for all runs	bf16 batching shifts teacher-forced log-probabilities by up to 1.09 nats; decided before the main run.
Validator v2/v3, *plain* style excluded, full equivalence screen, template styles	Audit acceptance below the pre-specified 85% threshold; decided before the main run.
Format stopping rule triggered (image removal only)	Attributable to model behavior (refusal), not to a pipeline error; forced-choice $P(\text{yes})$ retained, refusal reported, P1 and P3 re-estimated without refusals.
P2 computed on the model's own answer $\hat{y}$	The registered formula used the reference answer, which is unavailable at deployment and predicts correctness by construction.
P3 restricted to the test split	An analysis error had pooled the validation split; corrected to the registered set.
Chance correction; correctness and refusal decompositions; answer-change comparison across probes; per-dataset analysis	Added after the main run; descriptive, reported alongside the registered estimates.
Robustness probes R1–R3	Registered after the main run and before being run (pre-registration §15); R3 restricted to SLAKE as registered.
Additional pairs, H9–H11	Registered before being run (§16); all three new models passed the registered format check.
Probe R4, H12a–c	Registered before being run (§17), following the training-data review; H12a not supported, H12b reversed; under the registered decision rule, the hypothesis that the MedGemma models rely on the image more was withdrawn.
Training-data statements	Corrected after re-examining the documentation (§18): MedGemma-4B was trained on VQA-RAD; the MedGemma-1.5 model card lists only SLAKE.